Shao Boning

sbn252608@gmail.com ◇ +1 3417668056
sbnlight ◇ boning-shao.github.io ◇ boning-shao

EDUCATIONAL BACKGROUND

The University of Hong Kong — Hong Kong, China
BASc (AppliedAI) — 2023.09–Present

- CGPA: 3.65/4.3
- Relevant Coursework: Optimization (A+), Probability & Statistics (A+), Deep Learning (A).
- Honors: 2024–2025 Dean's Honours List.

University of California, Berkeley — Berkeley, CA, US
Berkeley Global Access Program — 2026.01–2026.08

- CGPA: 4.0/4.0
- Relevant Coursework: Advanced NLP, Bayesian Statistics, Numerical Analysis.

PUBLICATIONS

* Equal contribution; ‡ Project lead; † Equal advising. … denotes omitted authors.

Agentic World Action Models
Zhuoyang Liu*, Dantong Niu*†, Botong Zhang*, Boning Shao*, …,
Yuke Zhu†, Guanzhi Wang†, Linxi "Jim" Fan†, Trevor Darrell†.
ICLR, 2027. (Under review) [agentic-wam.github.io] [arXiv:2610.30136]

T-Rex: Tactile-Reactive Dexterous Manipulation
Dantong Niu*, Zhuoyang Liu*, Zekai Wang*, Boning Shao, …, Danfei Xu, Linxi "Jim" Fan, Trevor Darrell.
CoRL, 2026. [tactile-rex.github.io] [arXiv:2606.17055]

DexTacWAM: A Visuo-Tactile World-Action Model for Dexterous Manipulation
Haoran Yuan‡, Zekai Wang, Boning Shao, Haoran Lu, Trevor Darrell, Ismini Lourentzou†, Wei Zhan†.
arXiv preprint, 2026. [dextacwam.github.io] [arXiv:2609.24976]

Contrastive Action-Image Pre-training for Visuomotor Control (CAIP)
Yuvan Sharma*, Dantong Niu*‡, Anirudh Pai*, …, Boning Shao, …,
Jitendra Malik†, Trevor Darrell†, Roei Herzig†.
arXiv preprint, 2026. [caip-encoder.github.io] [arXiv:2606.17256]

Periodic Bootstrap Thompson Sampling for Periodically Non-Stationary Bandit Problems
Boning Shao.
DAI, 2025. [arXiv:2607.16986]

RESEARCH EXPERIENCE

AW-0: Agentic World Action Models — UC Berkeley
Undergraduate Researcher. Mentor: Trevor Darrell — 2026.06–2026.09

- Selected, preprocessed, and cleaned ~**2,000 hours of YAM robot data** and ~**1,000 hours of egocentric human data**, converting both sources into standardized **LeRobot** datasets for pretraining.
- Optimized the **code-labeling pipeline** and deployed large-scale labeling across the pretraining dataset.
- Built **evaluation infrastructure** for **2 embodiments**; trained, deployed, and evaluated **$\pi_{0.5}$, EgoScale, DreamZero, and WLA** baselines on **8 real-world** and **50 RoboTwin 2.0 tasks**.

T-Rex: Tactile-Reactive Dexterous Manipulation UC Berkeley
Undergraduate Researcher. Mentor: Trevor Darrell 2026.02–2026.05

- Designed **downstream manipulation tasks** for tactile-reactive policies.
- Collected ~**20 hours of midtraining data**, plus posttraining data for **6 downstream tasks**.
- Conducted **policy evaluations** on downstream manipulation tasks.

DataRank Platform Design Fudan University
Undergraduate Researcher. Mentor: Yanghua Xiao 2025.06–2025.08

- Built PageRank-inspired dataset search for DataRank, combining **semantic retrieval** with **BM25**.
- Developed interfaces connecting the algorithm to the database and frontend.

Multi-armed Bandit Algorithms Remote
Core member. Mentor: Osman Yagan, Carnegie Mellon University 2025.04–2025.07

- Implemented **Explore-Then-Commit, Upper Confidence Bound, and Thompson Sampling**, comparing their theoretical properties and performance across settings.
- Independently authored a paper on MAB enhancements, accepted for presentation at **DAI 2025**.

Model Benchmark Development and Evaluation The University of Hong Kong
Undergraduate Researcher. Mentor: Kai Han 2025.02–2025.03

- Annotated bounding boxes, attributes, interrelationships, and context for **30 images**; scored model outputs for **100 image sets** and helped analyze **10 models**.

PROJECT EXPERIENCE

RAG-Based Market Insight System for Startups The University of Hong Kong
Core Developer (Independent Module) 2025.04–2025.07

- Scraped Trustpilot reviews via **BeautifulSoup**; analyzed sentiment using **VADER** and star ratings.
- Built a **RAG** pipeline with **LangChain**, **ChromaDB**, and **Qwen-Turbo** to analyze reviews.
- Built a **Chainlit** UI to find market pain points and draft replies; tuned prompts to improve quality.

Patient Readmission Risk Prediction System Based on EHR The University of Hong Kong
Core Developer 2025.04–2025.07

- Built temporal and **TF-IDF** features from 171 variables and notes (**11,041 patients**; ~82,000 EHRs).
- Tuned **XGBoost, LightGBM, and CatBoost** with **Optuna**, using stratified 5-fold cross-validation for model selection and class weights to address class imbalance.
- Built a **rank-averaging ensemble**: **0.8146 validation AUC** and **0.82 Kaggle test AUC**.

INTERNSHIP EXPERIENCE

Infinigence AI Shanghai, China
Intern, Product Department 2024.07–2024.08

- Collected parameters, performance metrics, and pricing for **8 leading domestic and international AI models** (e.g., Qwen, Llama); contributed to a **10-page competitive analysis report**.
- Presented **Llama 3.1**'s technical report at a meeting, covering scaling laws and model training methods.

Moore Threads Shanghai, China
Intern, HSIO Department 2024.06–2024.07

- Learned the overall framework and processes of hardware systems and chip manufacturing systems.
- Refined a **Python script** to **categorize data** extracted from Excel, adapting it to updated chip design requirements and improving usability for the R&D team.

LANGUAGES AND SKILLS

- **Languages:** Mandarin (native); English (TOEFL 108/120).
- **Programming:** Python, C++, R.
- **Robotics:** Policy training, deployment, and evaluation for manipulation with grippers and dexterous hands; evaluation infrastructure development; data cleaning and preprocessing.
- **Machine Learning:** Transformers, diffusion models, CNNs; RAG; XGBoost, Optuna.
- **Computer Graphics:** OpenGL, 3D Gaussian Splatting.